

Transmission Rate

A text-first explanation of the essential physics behind link speed

Main message: a communication link has a finite transmission rate because the physical medium can only carry a limited range of signal changes, and the receiver must distinguish those changes from noise. Shannon's formula captures this upper physical limit.

Transmission delay = Packet size / Transmission rate

1. Start with the definition

Transmission rate tells us how quickly bits can be placed onto a link.

$$T_{\square} = L / R$$

Where:

$T_{\square}$ = transmission delay

L = packet length in bits

R = transmission rate in bits per second

The key point is that R is not just a software setting. It depends on what the physical channel can support. A link can only send data as fast as the medium, electronics, and signal quality allow.

- Higher R means lower transmission delay for the same packet size.
- Lower R means the packet takes longer to serialize onto the link.
- Distance affects propagation delay, not transmission delay directly.

2. Bits are represented by physical signals

Networks carry electromagnetic changes, not abstract symbols.

When a network sends data, the medium carries physical signals:

- Copper cable: changing voltages and currents.
- Fiber optic cable: light pulses.
- Wireless links: radio-frequency electromagnetic waves.

The receiver interprets these physical changes as symbols and then maps those symbols back to bits. Therefore, link speed depends on how quickly the medium can support distinguishable physical changes.

Simple wording: faster link speed means faster reliable signal changes.

3. Bandwidth means usable frequency range

A channel does not pass every possible signal change equally well.

Bandwidth is the range of frequencies a channel can carry usefully.

$$\text{Bandwidth} = f_{\text{high}} - f_{\text{low}}$$

Example: if a channel passes signals from 300 Hz to 3300 Hz, its bandwidth is 3000 Hz, or 3 kHz.

Why bandwidth matters for data:

- A bit stream changes over time.
- Faster changes require higher-frequency components.
- If high-frequency components are weakened, edges blur and symbols interfere.
- More usable bandwidth allows a potentially higher transmission rate.

4. How bandwidth is measured in practice

Engineers examine how the channel responds to different sine-wave frequencies.

A channel can be treated like a filter. To characterize it, engineers send test sine waves at different frequencies and measure the received strength. Low frequencies may pass well, while high frequencies may be attenuated or distorted.

- Send a low-frequency sine wave and measure the output.
- Increase the frequency and measure again.
- Plot received strength versus frequency.
- The usable part of that plot is the channel bandwidth.

Physical reason: resistance, capacitance, inductance, material loss, and interference make very rapid signal changes harder to preserve.

5. Noise limits how many states the receiver can distinguish

Bandwidth alone is not enough; the receiver also needs clarity.

Noise is unwanted variation added to the signal. Sources include thermal noise, interference, crosstalk, and imperfect electronics.

$$\text{SNR} = \text{Signal power} / \text{Noise power}$$

A higher signal-to-noise ratio means the intended signal stands out more clearly from random disturbance.

Why this affects transmission rate:

- With high SNR, the receiver can distinguish more reliable signal levels or symbols.
- With low SNR, symbols are more likely to be confused.
- Increasing signal power helps, but it has diminishing returns in Shannon's formula.

6. Shannon's formula: the physics limit

This formula combines bandwidth and noise into a maximum reliable data rate.

$$C = B \log_2(1 + S/N)$$

Meaning of the terms:

- C = channel capacity, the theoretical maximum reliable bit rate.
- B = bandwidth in hertz.
- S/N = signal-to-noise ratio.

What the formula teaches:

- More bandwidth increases capacity strongly.
- Better SNR also increases capacity.
- Because of the logarithm, simply increasing power has diminishing returns.

7. How this defines transmission rate

Real systems choose a usable rate below the physical maximum.

Physical medium → Bandwidth and noise → Shannon capacity C → Practical transmission rate R → Transmission delay $T_{\square} = L / R$

Transmission rate is the practical rate selected by the communication system. It must account for the physical channel, the allowed spectrum, the signal-to-noise ratio, the modulation/encoding method, and a margin for reliable operation. In ideal theory, R cannot exceed C . In real systems, R is usually chosen below C .

Short version: physics sets the ceiling; engineering chooses a safe operating speed below that ceiling.